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1 Dual-framework Testing of Species Distribution 2 Models for Predicting Elephant Habitat Suitability

3 **Abstract**

4 Automated machine learning (AutoML) offers transformative potential for model-
5 ing ecological trade-offs in wildlife conservation, particularly where management
6 interventions like fencing reduce human–wildlife conflict but fragment habitats
7 and undermine connectivity for keystone species. This case study evaluates the
8 impact of fence removal on the habitat suitability of six African elephants in
9 Kariega Game Reserve, South Africa. By coupling high-resolution telemetry
10 with fine-scale environmental data, the analysis highlight the potential for range
11 expansion and informs local management strategies.

12 We compared two modeling approaches: stacked ensemble species distribution
13 models (SSDMs) and the h2o AutoML framework, implementing density-based
14 adaptive thinning (DBSCAN) to mitigate spatial autocorrelation. Results demon-
15 strate that h2o AutoML significantly outperformed SSDMs in predictive accuracy
16 ($AUC \approx 0.90$ vs. 0.85), showcasing AutoML’s scalability in spatial ecology. Both
17 frameworks captured broad-scale habitat gradients with strong agreement (Pear-
18 son’s $r = 0.85\text{--}0.91$), yet AutoML revealed more pronounced post-fence expansion
19 dynamics through enhanced algorithmic flexibility. Pixel-wise difference maps
20 using Jaccard index effectively captured fine-grained spatio-temporal movement
21 patterns following fence removal.

22 AutoML predicted range expansion up to 35%, with core areas (top 25% suitable
23 pixels, $J_{0.75}$) showing substantial habitat gains. The top-ranked h2o AutoML
24 model was projected across climate scenarios (Shared Socioeconomic Pathways)

to assess future habitat shifts. This interdisciplinary approach demonstrates AutoML’s value for precise landscape delineation in keystone species conservation, providing an open framework and comprehensive benchmarking that complements established species distribution modeling pipelines while advancing conservation decision-making.

Keywords: fence removal, habitat suitability, ensemble modeling, h2o AutoML, machine learning

1 Introduction

Landscape restoration for the conservation of keystone species habitat is crucial for preserving ecological stability, biodiversity, and resilience (?). In an era of accelerating environmental change, their loss triggers cascading effects that fundamentally destabilize ecological networks and compromise essential ecosystem services (????). Many keystone species face escalating vulnerability to climate change, land-use change, including increasing human pressures (??) such as tourism, project development (?) and landscape modifications are not only restricting their movement but also altering behavioral patterns (??). Out of the many management interventions, fence removal stands central to the continuing debate on land sharing vs land sparing in human-wildlife coexistence (?). The most direct management challenge in southern Africa is wildlife fencing, established mainly to keep wildlife within reserves while boosting eco-tourism. Although boundary fences can substantially reduce human-wildlife conflict, internal fences may disrupt key ecological connectivity for wide-ranging animals like elephants (?????).

Particularly, keystone megafauna such as African elephants epitomize the tension between conservation and landscape modification for human well-being (??). Elephants serve as ecosystem engineers, influencing biodiversity and ecological processes via seed

50 dissemination, canopy alteration, soil and water management, and creating micro
51 habitats (??). Yet these functions are threatened by poaching, habitat fragmentation,
52 and climate change (?). Over the past fifty years, savanna elephant numbers have
53 dropped by around 70%, while forest elephant populations decreased by 90%, leading
54 to an estimated continental reduction of 77% (?). Simultaneously, atmospheric CO₂
55 concentrations have risen to 420 ppm, amplifying climate extremes and compounding
56 pressures on keystone taxa habitat (???). These elements make elephants not only a
57 conservation priority but also a sensitive indicator species for assessing how landscape
58 modifications and management interventions affect ecosystem integrity.

59 To quantify such changes, species distribution models (SDMs) provide a key
60 means of linking species' niche data with environmental variables, providing spatial
61 data for policymakers, reserve managers, and ecologists to assess habitat suitability,
62 predict spatial movements, and plan conservation strategies (??). High-resolution
63 remote-sensing products, such as Sentinel-2 and MODIS, now supply continuous
64 information on habitat data, vegetation, land cover, and human footprint. Leveraging
65 them markedly improves SDM predictive performance (?????). Integrating fine-scaled
66 species occurrence data with machine learning yields geographically refined forecasts
67 that inform proactive conservation and connectivity strategies (?).

68 Traditionally, SDMs utilize techniques such as maximum entropy (MaxEnt), random
69 forests, generalized linear models (GLMs), generalized additive models (GAMs), and
70 support vector machines (SVMs). Despite their efficacy, these algorithms require
71 substantial model calibration, parameter tuning, increased computational demand,
72 and are often prone to collinearity, sampling bias, and overfitting (?????). Ensemble
73 approaches, like those in the SSDM package, address underlying problems by integrating
74 multiple algorithms (??). Yet ensemble SDMs remain difficult to scale and standardize,
75 limiting reproducibility in applied contexts. Several studies (????) highlight the
76 absence of systematic benchmarking or reporting standards, making model comparison

77 and scalability challenging. Automated machine learning (AutoML) offers a scalable
78 alternative. By automating algorithm selection, hyperparameter optimization, and
79 ensemble integration, AutoML frameworks enhance reproducibility and reduce user bias
80 while maintaining predictive power (??). Early applications in engineering (??) indicate
81 AutoML can streamline ecological modeling (?), but systematic benchmarks against
82 established ensemble SDMs are scarce, especially for species-specific conservation
83 problems (??). We hypothesize that AutoML offers a more encouraging outlook with
84 stronger gains and lasting impact, while SSDM provides a more cautious, stability-
85 focused perspective. Given persistent challenges in SDM reproducibility and the absence
86 of standardized reporting protocols (??), open-access code and transparent workflows
87 are essential for enabling independent verification, methodological refinement, and
88 collaborative advancement in conservation science (??). Addressing this gap is critical
89 for assessing the credibility of AutoML-derived outputs and underscores methodological
90 integrity in forecasting dynamic ecological futures.

91 Relatively few studies have combined fine-scale telemetry with high-resolution
92 environmental predictors for elephants, despite the potential of such integration (???).
93 Telemetry data reveals movement and habitat use at the level of individual decisions,
94 while remote sensing provides continuous landscape gradients. Together, AutoML
95 enables fine-grained SDMs that link species behavior directly to dynamic environmental
96 conditions. In this study, we benchmark h2o AutoML (?) against traditional ensemble
97 SDMs using elephant telemetry data from Kariega Game Reserve, South Africa.
98 Specifically, we evaluate differences in predictive accuracy, computational efficiency,
99 and spatial predictions while examining how targeted fence removal influences habitat
100 suitability and potential range expansion. Since connectivity restoration represents a
101 long-term conservation commitment, we additionally project these models into future
102 climate scenarios (2021–2100) to assess the resilience of newly accessible habitats against
103 projected environmental shifts. Through this approach, we propose a reproducible,

104 scalable machine-learning framework that enhances predictive modeling for keystone
105 species and provides actionable insight for both immediate management and long-term
106 conservation planning ??.

107 2 Methods

108 2.1 Study Area

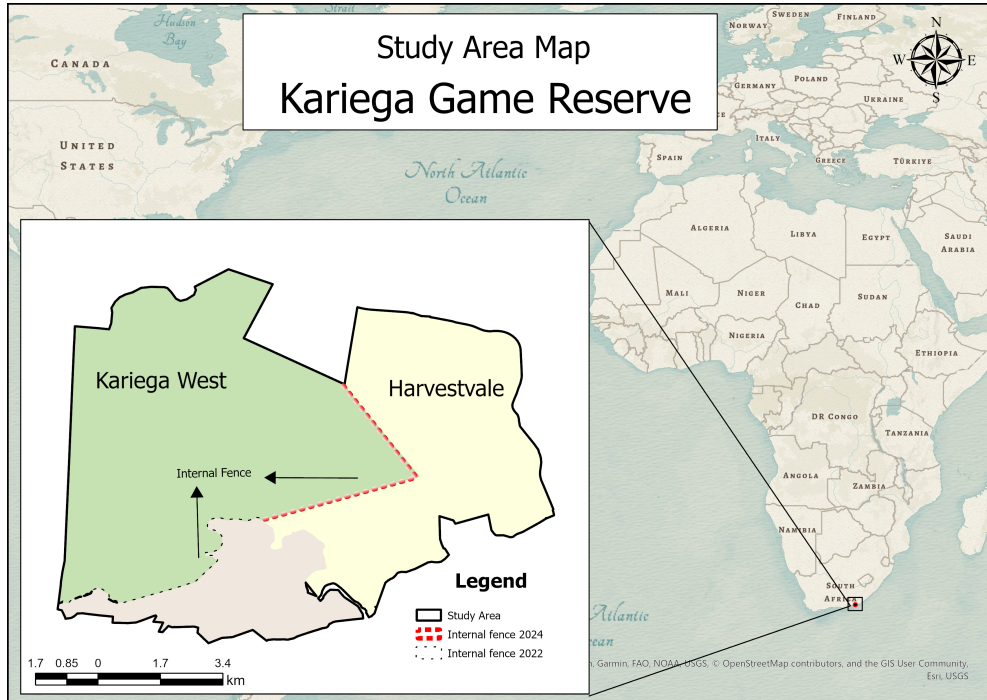


Fig. 1: Study Area Map.

109 The Kariega Game Reserve, located in the Eastern Cape province of South Africa,
110 is a premier private wildlife conservancy spanning 8,192 hectares (81.92 km²) along the
111 Bushmans River. The reserve lies 9.5 km inland from the Indian Ocean and is dominated
112 by dense, woody, and thorny semi-succulent vegetation characteristic of the Albany
113 Thicket biome. Kariega supports a diverse assemblage of wildlife, including African

elephant (*Loxodonta africana*), giraffe (*Giraffa camelopardalis*), cheetah (*Acinonyx jubatus*), white and black rhinoceros (*Ceratotherium simum* and *Diceros bicornis*), ostrich (*Struthio camelus*), and numerous avian and reptile species. At the start of this study, the elephant population (75 individuals) was divided by an internal fence into two sectors: Kariega West (KW; 2,792 ha; 55 elephants) and Harvestvale (HV; 4,369 ha; 20 elephants). Fence removal at Kariega occurred in two phases: in September 2022, HV was expanded by 1,031 ha following the first internal fence removal. A second removal in January 2024 eliminated the primary barrier between KW and HV, creating a contiguous 8,192 ha landscape accessible to the entire elephant population (?) (fig. 1).

2.2 Modelling Workflow

To ensure methodological rigour and reproducibility, we formalise the analytical process into a five-stage modelling pipeline (Fig. ?? and ??). This framework integrates diverse data streams into a single comparative diagnostic environment:

1. **Data Integration:** Harmonisation of 30-minute GPS telemetry with high-resolution (30 m) environmental co-variates (NDVI, Topography, Bioclimatic variables).
2. **Spatial Thinning:** Implementation of a density-based clustering algorithm (DBSCAN) (?) to mitigate spatial autocorrelation while preserving fine-scale movement intensity.
3. **Dual-Framework Training:** Simultaneous execution of h2o AutoML (leveraging hyperparameter-optimised ensembles) and traditional SSDM (leveraging classical algorithm stacking).
4. **Uncertainty Diagnostics:** Quantitative comparison of frameworks using Pearson's r , RMSE, and Jaccard overlap of suitability cores ($J_{0.75}$).
5. **Future Projection:** Extrapolation of validated models into multiple climate scenarios or Shared Socioeconomic Pathways (SSPs 2021–2100) to inform long-term habitat resilience strategy.

2.3 Data Collection and Pre-processing

Table 1: Dataset summary showing elephant IDs, record counts, temporal coverage before and after fence removal, and the number of points retained after DBSCAN spatial thinning.

Dataset	ID	Count	Earliest	Latest	DBSCAN
Before Jan 2024					
E1B	Kamva	12,364	09-01-2022 00:07	04-12-2023 23:40	1,106
E2B	Kambaku	10,364	09-01-2022 00:03	02-12-2023 23:40	927
E3B	Bukela	23,816	09-01-2022 00:07	31-12-2023 23:44	2,128
E4B	Half Moon	23,808	09-01-2022 00:23	31-12-2023 23:50	2,133
E5B	Beauty	18,629	09-01-2022 00:09	31-12-2023 23:56	1,671
E6B	Balu	21,259	09-02-2022 09:44	29-11-2023 17:45	1,901
After Jan 2024					
E3A	Bukela	25,091	01-01-2024 01:14	02-12-2025 23:54	2,246
E4A	Half Moon	24,999	01-01-2024 01:21	02-12-2025 23:56	2,244
E5A	Beauty	1,846	01-01-2024 01:57	14-02-2025 23:56	166
Total	–	162,175	–	–	14,522

To monitor spatial responses to habitat expansion, six African elephants (three solitary adult bulls and three matriarchs) were fitted with XL LoRa GPS collars (African Wildlife Tracking) in August 2022. Collars recorded locations every 30 minutes, enabling detailed analysis of daily movement patterns before and after fence removal. Data were transmitted remotely through the LoRa system and managed via EarthRanger software (?). GPS locations recorded outside of LoRa network coverage were stored in the collars’ internal memory and transmitted once connectivity was re-established, ensuring a complete spatial record of occupancy.

The telemetry dataset was subsequently cleaned and quality controlled. Duplicate timestamps and erroneous points were removed, and records were homogenized across individuals by standardizing coordinate reference systems (WGS84), date-time formats, and attribute schemas. The conclusive dataset encompassed a three-year duration, consisting of 127,717 GPS coordinates. Five datasets documented elephants before fence removal (09-01-2022 to 31-12-2023), whereas three monitored matriarchs after

155 fence removal (01-01-2024 to 02-12-2025) (Table. 1). Two matriarchs (E3 and E4)
156 originated from the Kariaga West (KW) section (55 elephants), while matriarch E5
157 was from the Harvesvale (HV) section (20 elephants). The GPS collar on E5 ("E5")
158 experienced device malfunction four months post-fence removal, yielding a smaller
159 dataset ($n = 1,121$ points) but still spanning a full year. Despite this, each file contained
160 sufficient records for robust habitat utilization maps and temporal species distribution
161 models (SDMs).

162 To account for the 30m resolution of environmental covariates (along with the
163 downsampled 1km resolution worldclim data) and mitigate spatial and temporal
164 autocorrelation, we implemented a density-based spatial thinning framework using the
165 DBSCAN algorithm (??). Unlike traditional grid-based thinning, DBSCAN explicitly
166 distinguishes between localized behavioral clusters (e.g., foraging or resting patches)
167 and exploratory 'noise' (?), allowing for sampling strategies that vary with local
168 point density. We parameterized the algorithm using a neighborhood radius (ϵ) of
169 0.004° ($\sim 400\text{m}$) and a minimum occupancy threshold ($minPts$) of 15 fixes. Given
170 our 30-minute sampling interval, this threshold represents ≥ 7.5 hours of localized
171 residency, ensuring that only significant behavioral patches were clustered for sampling.
172 Within identified clusters and noise points, we retained a 9% fraction ($\sim 14,500$ points
173 total 1) to reduce data redundancy while preserving the overarching geographic
174 footprint and density patterns of usage intensity. This strategy ensured that the final
175 training data reflect environmentally independent selections rather than temporal
176 streaks of movement, satisfying the assumption of spatial independence relative to our
177 environmental predictors (?).

178 Retaining local density is crucial for SDM predictive accuracy, as it preserves the
179 spatial structure of occurrence data relative to environmental covariates. Individual
180 telemetry data includes transient movements that may overestimate suitable habitat
181 (??). Inflated spatial autocorrelation, biased estimates, and reduced computational

182 efficiency are common when unthinned GPS data is used, especially when environ-
183 mental predictors are available at coarser resolutions than movement data. This was
184 particularly relevant for our high-frequency telemetry dataset (30-minute intervals)
185 with occasional irregularities due to network coverage gaps.

186 We implemented DBSCAN using the `dbscan` package in R with four parameters:
187 (i) neighbourhood radius (`eps_value` = 0.004 degrees, ~400 m); (ii) minimum points
188 per cluster (`minPts_value` = 15); (iii) fraction retained per cluster (`fraction_value`
189 = 0.09); and (iv) minimum samples for small clusters (`min_samples_value` = 4). This
190 strategy balanced down-sampling while retaining spatial heterogeneity and density pat-
191 terns. Thinning was applied separately to each individual (E1B, E2B, E3B, E4B, E5B,
192 E3A, E4A, E5A, E6B), maintaining core spatial structure while eliminating redundant
193 records and enhancing SDM robustness and computational efficiency (Fig. ??).

194 2.4 Environmental Data Preparation

195 We assembled 26 environmental predictors from multiple high-resolution sources. NDVI
196 and EVI were derived from Sentinel-2 imagery (10 m resolution) (?). Topographic
197 variables (elevation, slope, aspect) were extracted from SRTM DEM (30 m) (?).
198 Aspect was converted to radians to produce cosine-aspect ("northness") for Southern
199 Hemisphere orientation. Land cover was characterized using Global Forest Change ?
200 and ESA WorldCover products (10–30 m) (?). We incorporated 19 bioclimatic variables
201 (BIO1–BIO19) from WorldClim at 1 km resolution (?), representing annual trends,
202 seasonality, and extremes of temperature and precipitation (???). To ensure temporal
203 alignment, dynamic landscape predictors (NDVI, EVI, and LST) were period-specific:
204 2023 annual median composites were used for the pre-fence models (Set B), and 2024
205 median composites for the post-fence models (Set A). In contrast, bioclimatic variables
206 (BIO1–BIO19) while static topographic predictors were held constant across both
207 periods to represent the stable climatological and geomorphological baseline of the

208 Kariega landscape. Environmental predictors originated at varying native resolutions
 209 (10,m, 30,m, and 1,km). To ensure consistency across the modeling stack, all layers
 210 were aligned and resampled to a common 30,m spatial grid using bilinear interpolation
 211 for continuous variables and nearest-neighbor for categorical layers. This standardized
 212 resolution was chosen to balance the high-frequency telemetry data with the available
 213 topographic and vegetation co-variates. To reduce multicollinearity, we evaluated
 214 pairwise associations using Kendall's τ correlation (??), excluding highly correlated
 215 predictors ($r \geq 0.8$) via a correlation matrix ???. Later, we assessed variable importance
 216 for each predictor to examine the relative influence of climatic and topographic factors
 217 on the prediction of each species' presence. This particular metric is derived from the
 218 performance decline observed when each variable is sequentially excluded from the
 219 comprehensive model containing all variables (?).

220 **2.5 Species Distribution Modeling Frameworks**

221 **2.5.1 Stacked Ensemble SDM (SSDM)**

222 The ensemble SDM workflow was implemented using the SSDM R package (?). SSDM
 223 combines multiple algorithms including GLMs, GAMs, MARS, GBMs, CTAs, RFs,
 224 ANNs, and SVMs. Pseudoabsence points were generated internally to balance presence-
 225 only data, and models were trained using holdout cross-validation (75% training,
 226 25% testing). In the SSDM package, background points are generated internally by the
 227 package to optimize the discrimination between presence and landscape availability.
 228 Performance was evaluated with AUC, Cohen's κ (?), sensitivity, and specificity (??).
 229 By integrating multiple algorithms, SSDM ensembles reduce algorithm-specific biases
 230 and provide consensus predictions applicable to conservation planning (???).

231 2.5.2 h2o AutoML Workflow

232 We applied h2o AutoML using the h2o R package (??). Equal pseudoabsence points
233 ensured balanced training data. To generate the presence-absence datasets required for
234 the modelling frameworks, we utilized a fixed 1:1 ratio between presence (thinned GPS
235 fixes) and background (pseudo-absence) points, ensuring a balanced representation of
236 used versus available habitats. For the H2O AutoML framework, we further mitigated
237 sensitivity to background point selection by employing the `balance_classes` parameter
238 which is an internal rebalancing algorithm that uses SMOTE (Synthetic Minority
239 Over-sampling Technique) to optimize the discrimination between classes during the
240 training process. This dual-layered balancing strategy ensures that the suitability pre-
241 dictions are objective and independent of the initial sampling density. Eight individual
242 models captured fine-scale habitat use variation where AutoML automatically trained
243 base learners including GBMs, XGBoost, Deep Learning networks, DRF, and XRT,
244 integrating them via a meta-learner weighted by cross-validated AUC (?). Variable
245 importance was aggregated across ensembles. Spatial predictions included: (i) current
246 habitat suitability maps and (ii) future projections under four SSPs (SSP126, SSP245,
247 SSP370, SSP585) for periods 2021–2040, 2041–2060, 2061–2080, and 2081–2100.

248 2.6 Assessment and Validation

249 For both modeling frameworks, predictors were assembled into raster stacks, from
250 which point-level values were extracted at each presence and pseudo-absence location
251 for the respective elephant datasets. To ensure the credibility of our final predictions,
252 we then employed a multi-tiered validation approach following established protocols
253 and recommendations from (?????). This involved assessing framework consistency,
254 quantifying spatial agreement between AutoML and SSDM, and mapping temporal
255 transitions following fence removal.

2.6.1 Model Agreement Assessment Between Frameworks

Consistency between the automated and ensemble modeling pipelines was quantified through pixel-wise spatial correlation and error metrics. Predictive agreement was evaluated using Pearson's correlation coefficient (r), Spearman's rank correlation (ρ), root-mean-square error (RMSE), and mean absolute error (MAE):

$$r_{\text{Pearson}} = \frac{\sum_i (S_{\text{h2o},i} - \bar{S}_{\text{h2o}})(S_{\text{SSDM},i} - \bar{S}_{\text{SSDM}})}{\sqrt{\sum_i (S_{\text{h2o},i} - \bar{S}_{\text{h2o}})^2} \sqrt{\sum_i (S_{\text{SSDM},i} - \bar{S}_{\text{SSDM}})^2}}, \quad (1)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (S_{\text{h2o},i} - S_{\text{SSDM},i})^2}. \quad (2)$$

Where,

n is the total number of pixels evaluated within the Area of Interest (AOI).

i denotes the specific pixel index being compared.

Spatial difference maps ($\Delta_{\text{method}} = S_{\text{h2o}} - S_{\text{SSDM}}$) were generated to visualize areas of concordance and divergence. Furthermore, we quantified "hotspot" overlap using the Jaccard index ($J_{0.75}$), defined as the Intersection over Union (IoU) of the upper-quartile Suitability masks ($Q_{0.75}$) (?):

$$J_{0.75} = \frac{\sum (H_{\text{h2o}} \wedge H_{\text{SSDM}})}{\sum (H_{\text{h2o}} \vee H_{\text{SSDM}})}, \quad H = [S > Q_{0.75}(S)]. \quad (3)$$

Where,

S represents the predicted habitat suitability score ($[0, 1]$) for a given model.

H represents the binary hotspot mask derived from the suitability threshold.

By applying multiple thresholds ($Q_{0.25}$, $Q_{0.50}$, $Q_{0.75}$), we evaluated how model agreement and area change vary from broad potential range to optimal core habitat. This

analysis exposes methodological biases, highlighting how the choice of modeling pipeline influences conservation inferences under dynamic ecological and climatic conditions.

2.6.2 Replicates and Summary Layers

To assess internal model stability, three replicate models were fit for each individual $\times$ period $\times$ method combination. We derived per-pixel replicate summaries including the mean (μ), standard deviation (σ), and coefficient of variation ($CV = \sigma/(\mu + \varepsilon)$). We also calculated stability (ϕ_t) as the fraction of replicates exceeding a given suitability threshold (t). These diagnostics serve to identify regions of high predictive uncertainty and inform the subsequent spatial interpretation of the suitability maps.

2.6.3 Temporal Change Analysis (E3, E4, and E5)

For elephants with data spanning both periods, temporal change was evaluated using transition matrix methodology (?). Suitability shifts were quantified by the continuous difference $\Delta_{A-B} = S_A - S_B$, where positive values indicate habitat gain following fence removal. Categorical transitions (Gain, Stable, Loss) were summarized using logical operators at thresholds $t \in \{0.25, 0.50, 0.75\}$:

$$\text{Gain}_t = (A_t \wedge \neg B_t), \quad \text{Stable}_t = (A_t \wedge B_t), \quad \text{Loss}_t = (B_t \wedge \neg A_t). \quad (4)$$

Finally, we reported the % Area Change ($\%\Delta\text{Area}_t$) and temporal Jaccard similarity (J_t) to evaluate the magnitude and spatial persistence of suitability shifts. Gain/Stable/Loss (GSL) maps, Δ_{A-B} rasters, and Jaccard-vs-threshold curves provide threshold-explicit views of post-fence responses.

2.7 Reproducibility

All analyses were conducted in R (v4.3) using **terra**, **SSDM**, and **h2o**, with parallelization on a multi-core workstation. Rasters were aligned via re-projection, bilinear resampling for continuous layers and nearest-neighbor for binary outputs. Code paths, thresholds and file conventions are documented in the github repository (?) to ensure end-to-end reproducibility.

3 Results

3.1 Model discrimination and between-method agreement

Table 2: Between-method agreement metrics comparing h2o AutoML and SSDM predictions across all datasets.

Dataset	Pearson r	Spearman ρ	RMSE	MAE	Jaccard _{0.75}
E1B	0.952	0.926	0.096	0.058	0.787
E2B	0.928	0.910	0.115	0.066	0.772
E3B	0.947	0.911	0.110	0.066	0.789
E4B	0.946	0.922	0.112	0.069	0.796
E5B	0.936	0.917	0.102	0.068	0.680
E6B	0.929	0.916	0.108	0.072	0.684
E3A	0.878	0.886	0.113	0.088	0.597
E4A	0.839	0.840	0.170	0.133	0.577
E5A	0.856	0.861	0.206	0.143	0.597

Across eight telemetry datasets (E1B, E2B, E3B, E4B, E5B; E3A, E4A, E5A), we bench-marked h2o AutoML against stacked SDMs (SSDM). For elephants with before–after coverage (E3, E4, E5), temporal change was quantified as $\Delta_{A-B} = S_A - S_B$. Post-fence removal results indicated that reserve size and population density likely influenced habitat use, as two herds originated from the smaller, denser Kariaga West (KW) section and one from the larger Harvestvale (HV) section. Results include continuous suitability surfaces, threshold-ed maps ($Q_{0.25}$, $Q_{0.50}$, $Q_{0.75}$), and composite figures consolidating spatial predictions and summary metrics.

312 Both frameworks exhibited high correspondence in predicted suitability, yet with
 313 systematic differences in magnitude and spatial detail (Table. 2). Pixel-wise correlations
 314 were strong (Pearson's $r = 0.839\text{--}0.952$; Spearman's $\rho = 0.840\text{--}0.926$), indicating broad
 315 agreement in large-scale gradients. However, local discrepancies remained non-negligible
 316 (RMSE 0.096–0.206; MAE 0.058–0.143). Hotspot overlap, measured by Jaccard index
 317 on the upper quartile ($J_{0.75}$), was moderate (0.577–0.796), indicating substantial core-
 318 habitat agreement despite high global correlations. Jaccard values showed greater
 319 variability among matriarchs (0.684–0.787) than bulls, potentially reflecting increased
 320 habitat heterogeneity following fence removal. Spatial difference maps ($S_{\text{h2o}} - S_{\text{SSDM}}$)
 321 revealed that most divergences occurred along transitional zones and mixed-cover
 322 mosaics, where subtle environmental gradients amplify model-specific sensitivities.



Fig. 2: Hotspot Jaccard similarity (upper quartile, Q75) between h2o AutoML and SSDM predictions.

3.2 Environmental drivers of suitability

Both frameworks converged on precipitation and temperature as primary drivers, particularly precipitation seasonality (BIO15), precipitation of the warmest quarter (BIO16), and maximum temperature of the warmest month (BIO5). Post-fence datasets exhibited increased importance of vegetation productivity indices (NDVI and EVI). Rank correlation of variable importance between frameworks exceeded 0.85 (Fig. 3), indicating strong consistency despite spatial divergence in predictions.

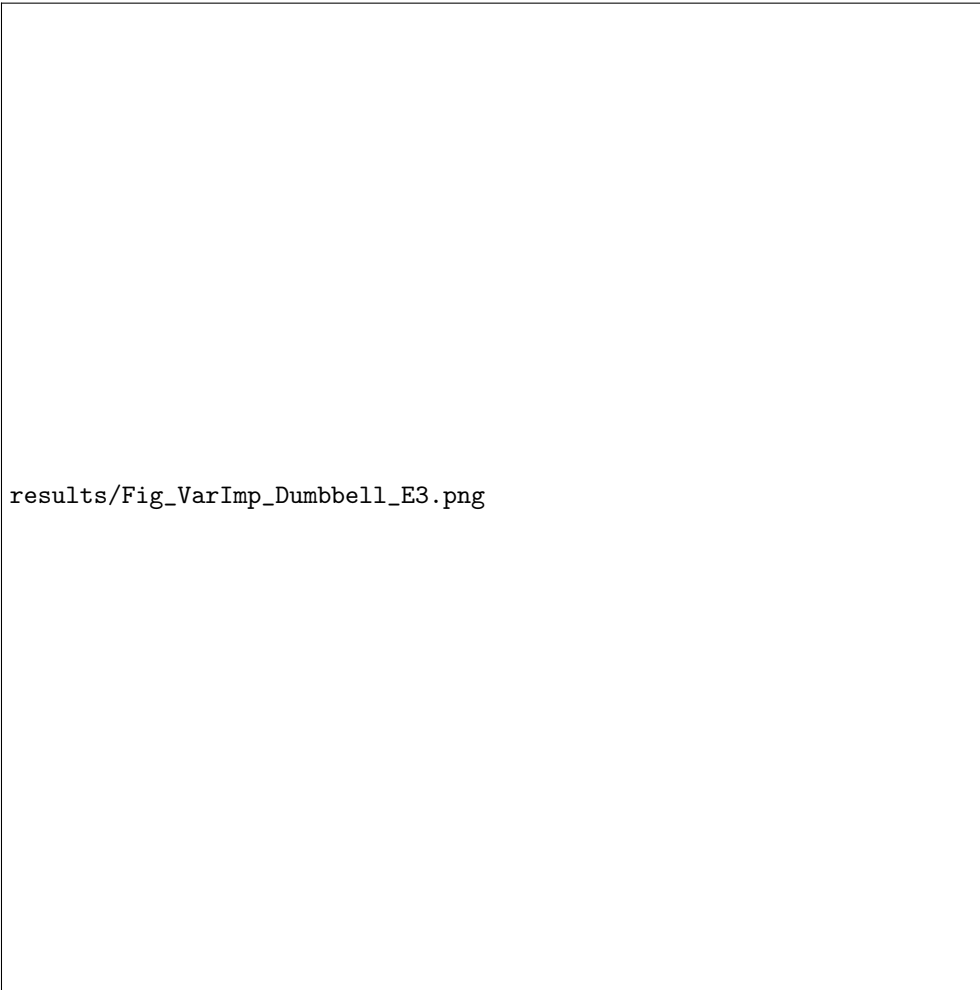


Fig. 3: Elephant E3 (“Bukela”). Dumbbell plots display the relative contribution (%) of each environmental predictor to h2o AutoML and SSDM habitat suitability models in the pre-fence removal period (gray circles; before 2024) and post-fence removal period (blue circles; after 2024). Horizontal line spans indicate the magnitude of change in predictor importance between periods, with leftward shifts representing decreased importance and rightward shifts representing increased importance. The distance between circles illustrates how fence removal altered the relative weight of environmental drivers in explaining habitat-use patterns. Variable importance rankings remained strongly correlated (Spearman’s $\rho > 0.85$) between frameworks, indicating agreement on dominant predictors despite differences in spatial detail. Figures for Elephant E4 (“E4”) and E5A (“E5”) are provided in the Appendix.

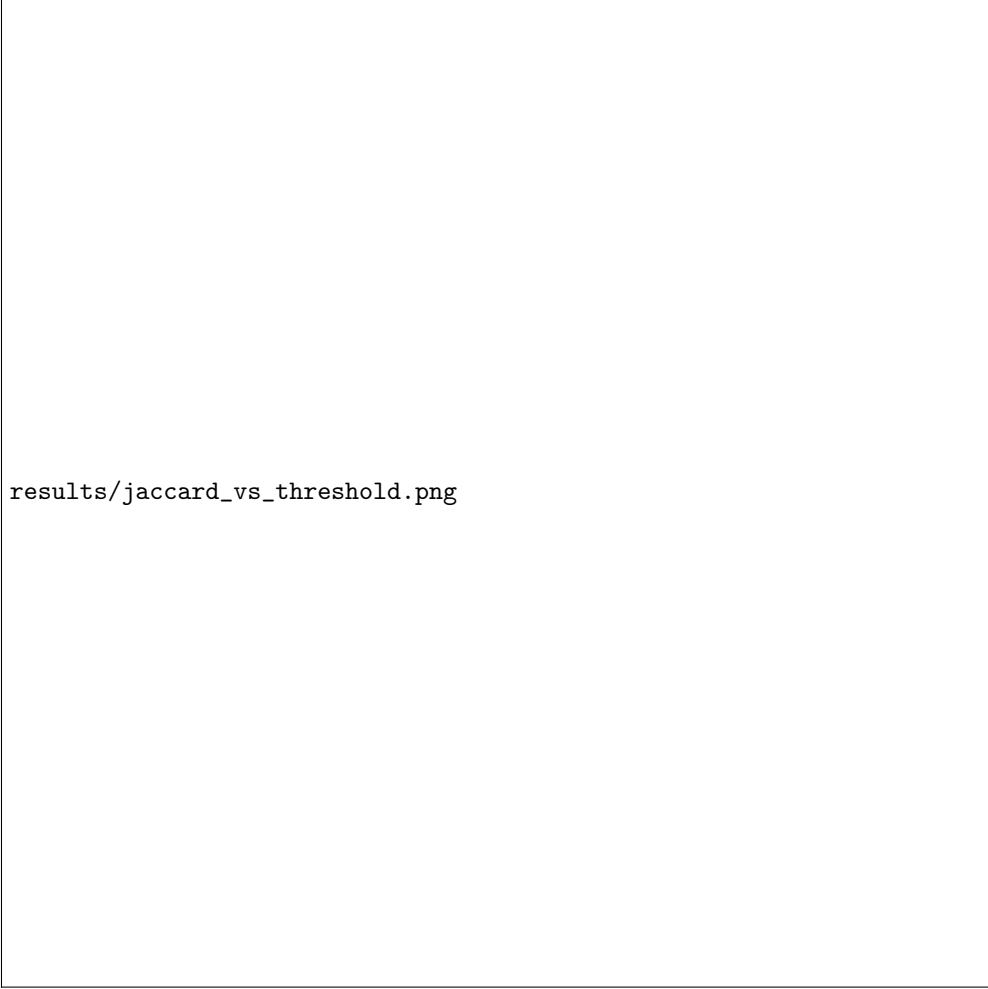
3.3 Temporal change following fence removal (E3, E4 & E5)

Table 3: Temporal change metrics (Before–After) for elephants E3, E4, and E5 across three suitability thresholds ($t \in \{0.25, 0.5, 0.75\}$) and two modelling frameworks (SSDM and h2o AutoML). Threshold t is the suitability cutoff. B_{area} and A_{area} are suitable habitat areas *Before* and *After* fence removal, respectively; $\% \Delta \text{ Area} = \frac{A_{\text{area}} - B_{\text{area}}}{B_{\text{area}}} \times 100\%$. J_t is the Jaccard similarity between Before and After maps (0–1). All areas are in km^2 .

ID	Method	t	B_{area} (km^2)	A_{area} (km^2)	Loss (km^2)	Stable (km^2)	Gain (km^2)	$\% \Delta$ (%)	J_t
E3	SSDM	0.25	30.13	51.82	17.78	12.35	39.47	71.98	0.18
E3	SSDM	0.50	18.28	25.12	15.76	2.51	22.60	37.43	0.06
E3	SSDM	0.75	12.87	6.41	12.74	0.13	6.28	−50.18	0.01
E3	h2o	0.25	59.38	86.92	14.73	44.65	42.27	46.37	0.44
E3	h2o	0.50	18.36	61.44	16.28	2.08	59.36	234.59	0.03
E3	h2o	0.75	11.47	39.74	11.32	0.15	39.60	246.56	0.00
E4	SSDM	0.25	30.53	49.63	24.89	5.64	43.99	62.55	0.08
E4	SSDM	0.50	19.05	31.42	18.92	0.13	31.29	64.92	0.00
E4	SSDM	0.75	13.27	17.61	13.27	0.00	17.61	32.64	0.00
E4	h2o	0.25	59.80	80.07	19.97	39.84	40.24	33.90	0.40
E4	h2o	0.50	53.87	61.35	18.55	35.32	26.02	13.88	0.44
E4	h2o	0.75	10.87	42.83	10.76	0.11	42.72	294.16	0.00
E5	SSDM	0.25	35.35	24.40	11.11	24.25	0.15	−30.98	0.68
E5	SSDM	0.50	19.89	12.89	10.17	9.73	3.17	−35.19	0.42
E5	SSDM	0.75	8.53	1.40	7.36	1.17	0.23	−83.63	0.13
E5	h2o	0.25	67.67	70.76	4.34	63.34	7.42	4.56	0.84
E5	h2o	0.50	52.62	59.18	4.81	47.81	11.37	12.48	0.75
E5	h2o	0.75	42.88	45.64	3.85	39.03	6.62	6.45	0.79

Both frameworks revealed strong threshold-dependent differences in habitat area and temporal change (Table 3), though with systematic framework divergence. At the permissive threshold ($t = 0.25$), both predicted substantial habitat expansion following fence removal (SSDM: up to 51 km^2 ; h2o: up to 86 km^2), indicating broad agreement on post-removal habitat gain. As thresholds became stricter ($t = 0.50, 0.75$), suitable area decreased monotonically and framework divergence increased markedly. SSDM predicted substantial habitat loss for E3 ("E3", −50%) and E4 ("E4", 32.64%) at $J_t = 0.75$, with Jaccard indices of 0.01 indicating minimal temporal

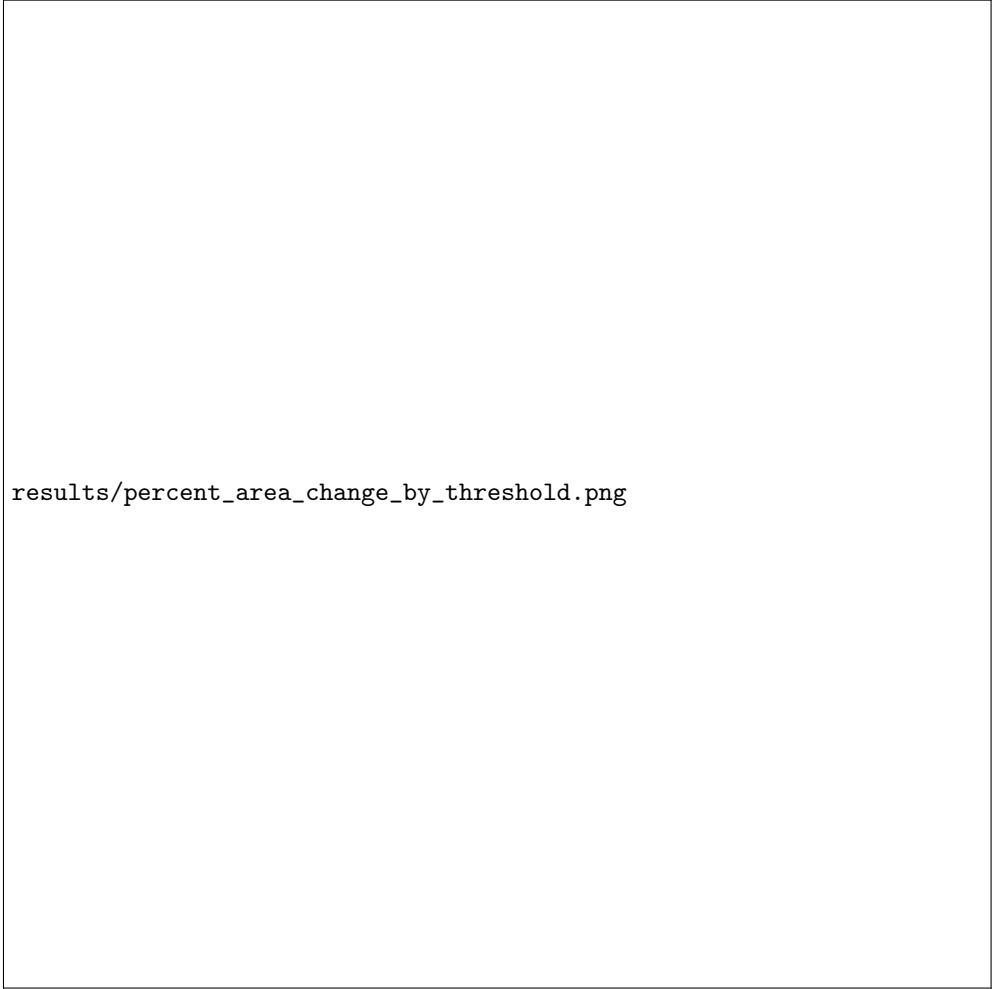
339 overlap of high-suitability cores. In contrast, h2o retained substantial core habitat
340 at the strictest threshold (Jaccard up to 0.79 for E5A), indicating greater predicted
341 persistence. For E5A ("E5"), frameworks diverged most sharply: SSDM predicted
342 progressive contraction across all thresholds (-30% to -83%), whereas h2o identified
343 net persistence despite modest area reduction. These differences highlight how ensemble
344 structures and algorithms influence temporal habitat projections.



results/jaccard_vs_threshold.png

Fig. 4: Jaccard(A,B) vs. threshold. Temporal robustness: overlap across thresholds (0.25, 0.50, 0.75) for E3, E4 and E5 within each framework.

345 Jaccard similarity indices (J_t) quantified spatial overlap between Before and After
 346 habitat maps at three thresholds ($t = 0.25, 0.50, 0.75$). h2o AutoML consistently
 347 showed higher temporal overlap than SSDM, particularly at moderate thresholds. For
 348 instance, $J_{0.25}$ averaged 0.44–0.84 for h2o but only 0.08–0.68 for SSDM, indicating more
 349 temporally coherent suitability patterns. At $t = 0.75$, overlap declined sharply for SSDM
 350 (often $J_t = 0.01$), reflecting minimal persistence of high-suitability zones. In contrast,
 351 h2o exhibited moderate to high persistence (E3: $J_{0.75} = 0.00$; E4: $J_{0.75} = 0.002$;
 352 E5: $J_{0.75} = 0.79$), with E5 showing notably stable core habitats ($A = 42.88 \text{ km}^2$).
 353 Composite panels integrate base maps, Δ maps, gain–stability–loss (GSL) strips, and
 354 metric tables (Figs 6, 7, ??, ??).



results/percent_area_change_by_threshold.png

Fig. 5: Percent area change (A vs. B) by threshold. Temporal robustness: area change across thresholds (0.25, 0.50, 0.75) for E3, E4 and E5 within each framework.



Fig. 6: E3 - h2o AutoML

355 **3.4 Synthesis**

356 We observe three consistent patterns. (i) *Discrimination*: h2o AutoML outperformed
357 SSDM across all datasets, achieving lower error and higher AUC. (ii) *Spatial agree-*
358 *ment*: Global correlations were high, but hotspot overlap remained moderate and
359 spatial difference maps revealed fine-scale divergence, particularly in transitional zones.
360 (iii) *Temporal change*: Both frameworks detected post-fence habitat expansion, with

361 h2o offering a more sensitive lens (sharper Δ , larger gains) and SSDM providing
362 a conservative baseline (smoother cores, higher stability). Together, these perspec-
363 tives delineate the plausible uncertainty envelope for conservation interpretation and
364 highlight the complementary insights of automated versus ensemble SDM approaches.

results/E3_SSDM_before_after_panel.png

Fig. 7: E3 - SSDM Per-elephant comparison for E3 (E3). The upper panel shows habitat suitability maps before and after fence removal, with a difference map $\Delta = A - B$ indicating increases (red) and decreases (blue) in suitability. The lower panel presents threshold-specific spatial changes: blue denotes areas that became suitable (Gain), red indicates areas that became unsuitable (Loss), and white shows areas that remained suitable (Stable) after the intervention. The accompanying table summarizes model metrics, Jaccard similarity (spatial overlap at each threshold) and percentage change in suitable area ($\% \Delta$ area). All areas are in km^2 .

4 Discussion

This study demonstrates that h2o AutoML consistently outperformed traditional SSDM across multiple performance metrics. AutoML achieved superior discrimination (higher ROC–AUC), calibration (lower RMSE and log-loss), and computational efficiency, revealing stronger capacity to capture nonlinear environmental responses (??). Unlike traditional ensemble approaches (?) using fixed weighting, AutoML adaptively reweighted base learners to match each elephant’s movement data structure. GBM- and XRT-dominated ensembles captured E3’s patterns, while E4 required deeper neural networks, and E5’s smaller sample was represented by simpler GBM architecture. AutoML’s robustness maintained high predictive accuracy, adding rigor to future climatic extrapolations (Fig .11) and aligning with findings regarding the framework’s robustness in other ecological applications (?). The distinction between structural connectivity (the removal of the fence) and functional connectivity is central to our findings. While traditional connectivity models often focus on potential movement pathways, our SDM approach provides an empirical record of habitability outcomes: it defines ‘where they were vs. where they could be’ by mapping the expansion of environmental suitability. These findings align with evidence that machine learning automation enhances predictive power in wildlife conservation (??), while its parallelized architecture (?) substantially reduced training time.

Addressing collinearity is fundamental for reliable inference (??). While some perspectives suggest that collinearity is not necessarily a ‘disease’ to be cured but an inherent feature of complex ecological datasets (??), we nonetheless employed Kendall’s τ as a robust screening tool for variable selection. This approach allowed us to identify the most ecologically interpretable variables while acknowledging that predictors in wildlife systems are intrinsically linked (?). Some studies have addressed collinearity through Variance Inflation Factor (VIF) and Principal Component Analysis (PCA), though both methods possess recognized limitations: VIF can be highly sensitive to

Table 4: Model performance comparison: AUC scores for H2O AutoML and SSDM.

Elephant	Period	H2O AutoML (AUC)	SSDM (AUC)
E1	Pre (B)	0.980*	0.875
E2	Pre (B)	0.987*	0.905
E3	Pre (B)	0.957*	0.880
	Post (A)	0.759^a	0.766
E4	Pre (B)	0.961*	0.874
	Post (A)	0.774^a	0.760
E5	Pre (B)	0.945*	0.852
	Post (A)	0.834^a	0.843
E6	Pre (B)	0.942*	0.858

Notes: ^a Post period H2O AutoML uses 5-fold CV performance. * Pre period H2O AutoML uses internal training fit (in-sample). Bold values highlight convergence of predictive accuracy post-fence removal.

sample size and scaling thresholds (??), while PCA often obscures ecological meaning by collapsing raw predictors into abstract, uninterpretable components (?). Spatial block cross-validation (??) confirmed predictive stability across spatially independent partitions (Figs. ??, ??). By pre-screening variables for ecological interpretability, AutoML focused automated feature selection on biologically meaningful predictors (??), balancing accuracy with transparency which is critical for conservation decision-making (?).

Pre- and post-fence outputs (Table. 3) provide compelling evidence of restoring landscape connectivity value. Before removal, elephants occupied restricted, fragmented habitats; post-removal revealed clear expansions into newly accessible terrain, mirroring broader evidence that fences constrain movement and erode long-term viability (??). Site-specific differences in reserve configuration shaped initial habitat use, with localized population densities (KW: 55 elephants; HV: 20 elephants) and spatial heterogeneity mediating connectivity gains (?).

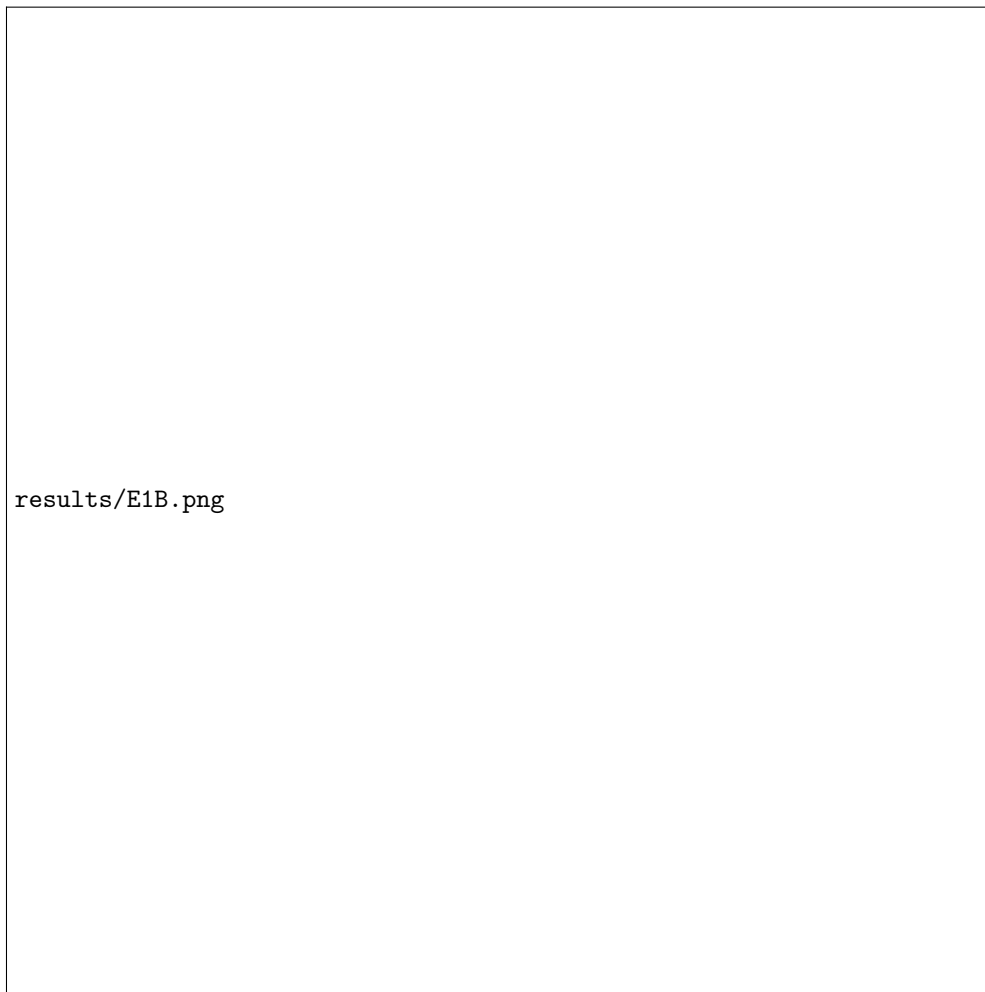


Fig. 8: Pre-fence removal habitat suitability map for male elephant E1B (“Kambaku”).

406 Individual responses revealed significant habitat expansion E3 and E4 (Fig. 4).
407 However, peculiar pattern was observed in E5 (Fig. 5) and the reason being the
408 failure of the telemetry device providing intermittent and incomplete GPS records.
409 Matriarch E3 (E3A) demonstrated the most pronounced expansion (Fig. 7), suggesting
410 experienced individuals effectively identify and utilize restored connectivity. AutoML
411 maintained stronger temporal consistency for E3, preserving continuous high-suitability
412 patches under strict thresholds, while SSDM fragmented predicted areas. E4 (E4)

413 displayed heterogeneous patterns with sharp gain-loss mosaics, reflecting exploratory
414 use typical of post-barrier investigations (??). E5 (E5) showed modest but coherent
415 expansion along existing corridors and retaining stable cores. Although the collar
416 eventually malfunctioned but with the available predictions it suggests her herd had
417 established optimal habitat selection with movement patterns (??). Expansion for both
418 E3 and E4 was directed toward Harvestvale, indicating restored connectivity enabled
419 access to previously inaccessible high-quality patches. These patterns highlight the
420 importance of assessing habitat change across multiple thresholds to identify expansion
421 fronts and persistent niche (??).

422 Despite AutoML’s superior accuracy, it exhibited limited spatial congruence with
423 SSDM, sharing only 17–28% of high-suitability areas, consistent with benchmarking
424 studies showing algorithmic ensembles often differ spatially despite comparable dis-
425 crimination scores (??). Difference maps ($\Delta = S_{\text{AutoML}} - S_{\text{SSDM}}$) reveal h2o AutoML
426 precisely delineated fence boundaries with sharp transitions corresponding to physical
427 barrier locations, while SSDM provided moderate spatial separation. This high-
428 lights AutoML’s enhanced sensitivity to complex ecological discontinuities and ability
429 to capture fine-scale habitat boundaries critical for conservation planning. Cross-
430 framework comparisons expose predictive confidence boundaries, underscoring the value
431 of ensemble approaches incorporating multiple algorithmic perspectives for high-stakes
432 conservation decisions (??).

433 Precipitation and temperature variables (Figs. 3, ??, ??) emerged as dominant
434 predictors, consistent with water-dependent megafauna ecology requiring reliable
435 surface water and thermally suitable foraging areas (??). Following fence removal,
436 vegetation indices (NDVI, EVI) gained prominence, indicating elephants rapidly
437 exploited newly accessible, high-productivity areas, suggesting quality forage became
438 more influential once barriers were removed (?).

439 Forward projections revealed contrasting trajectories through 2100. Under low-
440 emission scenarios (SSP1–2.6), suitable habitat remained stable, suggesting current
441 conservation areas may continue supporting populations. However, high-emission
442 pathways (SSP5–8.5) predicted substantial habitat loss, consistent with projected
443 climate niche contractions (??). These projections emphasize the critical importance of
444 maintaining and expanding connectivity now, as gains achieved through fence removal
445 may prove essential for enabling climate-driven range shifts.

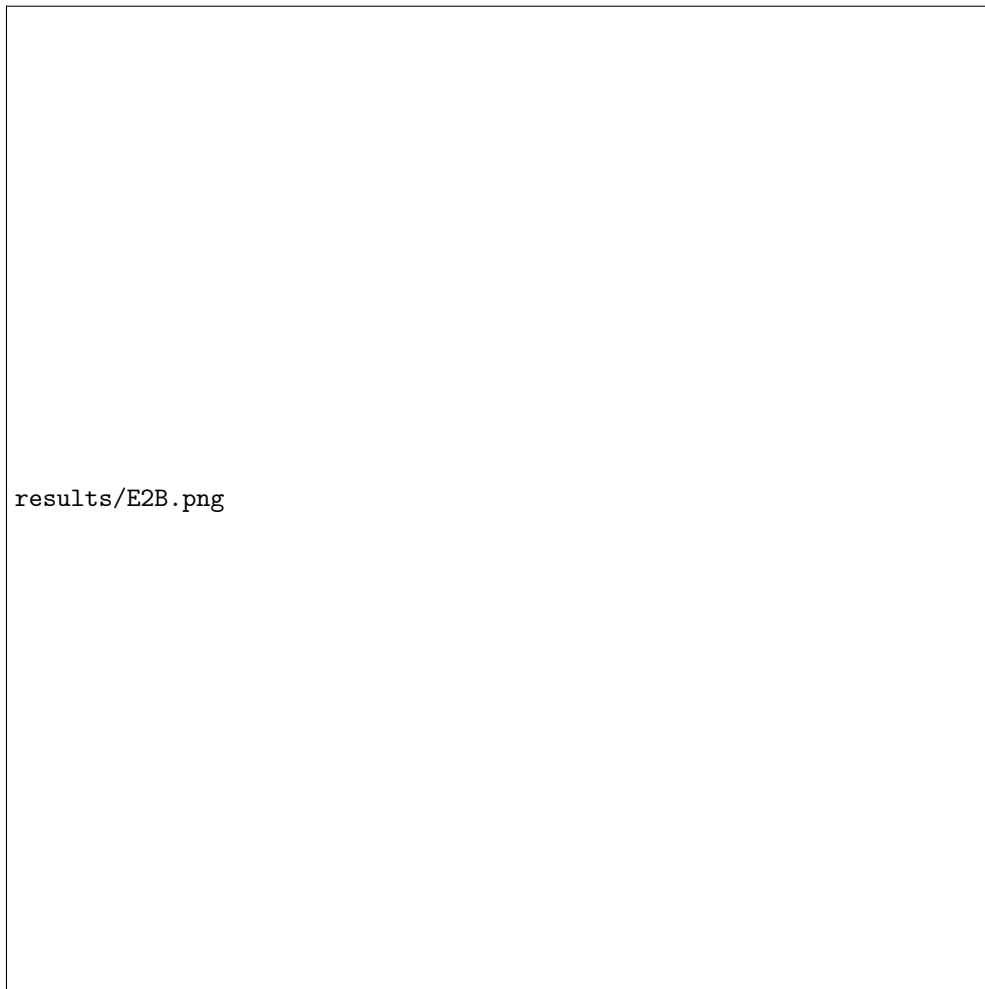


Fig. 9: Pre-fence removal habitat suitability map for male elephant E2B (“Kamva”).

446 Several limitations constrain broader generalization. After initial fence removal, col-
 447 lared bulls concentrated within Harvestvale while E6B explored widely (Fig. 8–10), and
 448 suitability maps revealed persistent patterns along remaining internal fences, indicating
 449 incomplete connectivity restoration. While this study focuses on modeling habitat suit-
 450 ability via SDMs, our results provide an empirical measure of functional connectivity
 451 restoration. By documenting the spatial expansion of high-suitability patches into pre-
 452 viously fenced sectors (particularly for E3 and E4), we demonstrate that the removal
 453 of structural barriers directly allowed for the realization of previously latent habitat
 454 potential. These findings provide a quantitative baseline for the connectivity gains
 455 achieved, which we further explore through mechanistic movement analysis in a com-
 456plementary study. Only complete fence removal enabled substantial habitat expansions,
 457 suggesting incremental improvements may be insufficient. Individual-level variation as
 458 E3’s extensive utilization versus E5’s corridor-following suggests conservation strate-
 459 gies must account for behavioral diversity. All three bulls experienced collar twisting
 460 within six months, necessitating removal and losing male movement data post-2024
 461 (?). Small sample size (six elephants, reduced to three post-fence) restricts understand-
 462 ing of sex-specific responses. Correlative models cannot fully capture behavioral or
 463 physiological processes underlying habitat selection (??), and short-term disturbances
 464 (droughts, fires, anthropogenic pressures) remain inadequately represented with only
 465 abiotic predictors. Future research should integrate mechanistic approaches (Step
 466 Selection Functions) with AutoML frameworks to capture biological realism (???).
 467 Dynamic connectivity models (??) and energetically informed dispersal kernels (?)
 468 could provide realistic representations of movement costs. Incorporating multimodal
 469 datasets (?) such as remote sensing time series, acoustic monitoring, genetic sampling,
 470 human–wildlife conflict data, could enable comprehensive predictions (???). AutoML’s
 471 computational efficiency suggests significant potential for scaling to landscape, regional,
 472 or continental analyses. Embracing open science principles through open-source code

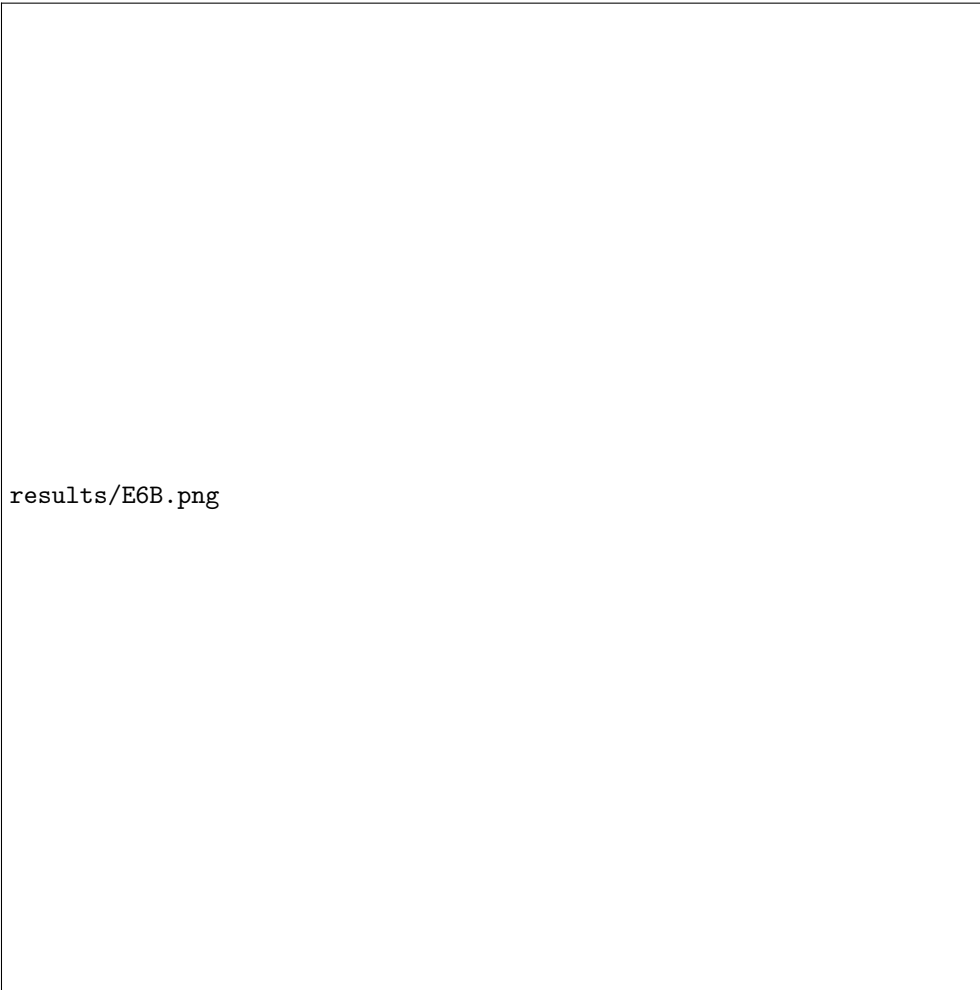


Fig. 10: Pre-fence removal habitat suitability map for male elephant E6B (“Balu”).

473 sharing, bench-marked validation protocols, and transparent reporting (??) is critical
474 to advancement.



Fig. 11: Future projection for E3-E3A.

475 Practical conservation implications are immediate and urgent. Quantified habi-
476 tat gains following fence removal (Table. 3) provide strong empirical support for
477 connectivity restoration. Our open-sourced code (?) leverages AutoML’s precise delin-
478 eation of expansion areas to inform targeted protection efforts, anti-poaching patrol
479 routes, and human–wildlife conflict mitigation in newly utilized habitats. Climate
480 projections (Fig. 11) indicating substantial habitat loss under high-emission scenarios
481 emphasize urgency of current connectivity restoration efforts (??). Spatial divergence

482 between AutoML and SSDM highlights the importance of using multiple approaches
483 for high-stakes conservation decisions, with management plans incorporating uncer-
484 tainty estimates across modeling approaches. By combining AutoML’s scalability with
485 mechanistic ecological understanding and multi-modal data integration, future research
486 can advance toward adaptive, data-rich conservation strategies reflecting ecological
487 complexity and practical management constraints while maintaining transparency and
488 reproducibility essential for science-based conservation policy.

489 5 Conclusion

490 This study therefore demonstrates that AutoML represents a significant advancement
491 in habitat suitability modeling, offering superior predictive performance, computa-
492 tional efficiency, and replicability compared to traditional ensemble approaches. The
493 framework’s adaptive optimization and ecological understanding makes it particularly
494 valuable for species-specific conservation planning. Our quantified benefits of fence
495 removal provide compelling evidence that landscape connectivity restoration enables
496 rapid exploitation of newly accessible, high-quality habitats by individual elephants.
497 However, spatial divergence between modeling frameworks and projected climate
498 impacts underscore the need for adaptive management that decomposes uncertainty
499 across data-driven (telemetry and thinning), algorithmic (AutoML vs. SSDM), and
500 temporal dimensions while strengthening large-scale restoration policies.

501 **Key Take-Home Messages:**

- 502 1. **AutoML as a conservation tool:** Integrating automated machine learning
503 with open-source frameworks and multi-modal datasets provides a scalable,
504 policy-relevant pathway for robust conservation planning, enabling more effective
505 biodiversity protection in an increasingly complex world.

506 **2. Connectivity restoration works:** Fence removal success in this system offers
507 a replicable template for global connectivity restoration efforts, demonstrating
508 how cutting-edge computational approaches can deliver immediate management
509 guidance while building adaptive, evidence-based conservation strategies.